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INTERNSHIP REPORT

IMPLEMENTATION OF A NETWORK INTRUSION DETECTION SYSTEM(NIDS)

CASE STUDY: AXA CAMEROON agence EMERAUDE.

A research and analysis project submitted to JFN HIGHTECH UNIVERSITY INSTITUTE in partial fulfillment to obtain an award of a PROFESSIONAL BACHELOR.

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ACADEMIC YEARS: 2025-2026

DESIGN AND IMPLIMENTATION OF A NIDS.



CERTIFICATION

This is to certify that this internship report entitled IMPLEMENTATION OF A NETWORK INTRUSION DETECTION SYSTEM (NIDS) was originally written by ESHON BRADLEY ADRIANO to meet the partial requirements and the regulations governing the award of PROFESSIONAL BACHELOR DEGREE.

Mr. NDZO ROLAND

Mrs. DAMLEPIN YOUNSIE KERINE

DATE: _____

DATE: _____

SIGNATURE: _____

SIGNATURE: _____

DECLARATION

I ESHON BRADLEY ADRIANO declare that I am the sole author of this report. I understand the nature of plagiarism, and I am aware of the institution's policy on this. I declare that this is an original work by me and contains neither material that has been accepted for the award of a certificate in any other institution, except where due acknowledgements have been made.

Date:

Signature:

DEDICATION

**TO THE ESHON'
FAMILY**

ACKNOWLEDGEMENT

This piece of work has been difficult and full with a lot of obstacles but by God's grace, I was able to accomplish the task. I give thanks to God almighty who is the mastermind and overall support system throughout this project.

I also give thanks to all the people around me who supported and encouraged me in one way or the other;

- ❖ Mr. ALPHONSE NAFACK: The proprietor of JFN for providing us with the resources and infrastructure to carry out our project.
- ❖ Mr. NDZO ROLAND: My academic supervisor for technical lectures he gave me which has contributed a lot in the realization and completion of this project.
- ❖ Mrs. DAMLEPIN YOUMSIE KERINE: My professional supervisor for the skillful training she gave me which has contributed a lot in the realization and completion of this project.
- ❖ To my parents: for all their unwavering support and encouragement throughout this project.
- ❖ To all my lecturers and the entire staff of JFN.
- ❖ To my friends for their guidance and precious advice throughout the year.
- ❖ To my entire family for their endless prayers and support.



ABSTRACT

The **Intrusion Detection System (IDS)** is a network security solution designed to operate within a virtualized lab environment, facilitating real-time monitoring and threat detection across a simulated enterprise network. The system allows security administrators to **monitor incoming and outgoing network traffic, detect malicious activity, and receive automated alerts**, while the IDS engine continuously analyzes packet data against predefined detection rules. This project aims to digitize and automate the threat detection process by integrating a structured rule-based system that efficiently handles **network traffic analysis, attack signature matching, and alert logging**.

The idea to develop this IDS was inspired by a professional observation made during an internship at **AXA Cameroon**. While working within the IT department, I noticed several security gaps, including the absence of real-time network monitoring, no automated alert mechanism, and a complete lack of visibility into suspicious network activity, which often left the infrastructure exposed to undetected threats and reconnaissance attempts. Recognizing the growing frequency of cyberattacks targeting enterprise networks and the increasing reliance on digital infrastructure for sensitive business operations, I saw an opportunity to create a more **efficient, proactive, and scalable** network security monitoring solution.

With this in mind, we embarked on the implementation of a fully functional Network-based Intrusion Detection System that integrates **custom detection rules, automated alert generation, and a wider range of attack scenario coverage**. This system eliminates the blind spots of unmonitored network infrastructure by providing a centralized monitoring platform where administrators can **detect, log, and respond to threats** effortlessly. In addition, I used various tools to realize the project, some of which are **VirtualBox** and **Ubuntu Server** for the virtual environment, **Snort** for intrusion detection, **Kali Linux** (Nmap, Hydra, hping3) for attack simulation.

RESUME

Le **Système de Détection d'Intrusion (IDS)** est une solution de sécurité réseau conçue pour fonctionner au sein d'un environnement de laboratoire virtualisé, facilitant la surveillance en temps réel et la détection des menaces à travers un réseau d'entreprise simulé. Le système permet aux administrateurs de sécurité de **surveiller le trafic réseau entrant et sortant, de détecter les activités malveillantes et de recevoir des alertes automatisées**, tandis que le moteur IDS analyse en continu les données des paquets par rapport à des règles de détection prédéfinies. Ce projet vise à numériser et à automatiser le processus de détection des menaces en intégrant un système structuré basé sur des règles qui gère efficacement **l'analyse du trafic réseau, la correspondance des signatures d'attaque et la journalisation des alertes**.

L'idée de développer cet IDS est née d'une observation professionnelle faite lors d'un stage à **AXA Cameroun**. En travaillant au sein du département informatique, j'ai remarqué plusieurs failles de sécurité, notamment l'absence de surveillance réseau en temps réel, l'absence de mécanisme d'alerte automatisé et un manque total de visibilité sur les activités réseau suspectes, ce qui laissait souvent l'infrastructure exposée à des menaces et des tentatives de reconnaissance non détectées. Conscient de la fréquence croissante des cyberattaques ciblant les réseaux d'entreprise et de la dépendance grandissante à l'infrastructure numérique pour les opérations commerciales sensibles, j'ai vu là une opportunité de créer une solution de surveillance de la sécurité réseau plus **efficace, proactive et évolutive**.

C'est dans cette optique que nous nous sommes lancé dans la mise en œuvre d'un Système de Détection d'Intrusion basé sur le réseau, pleinement fonctionnel, **intégrant des règles de détection personnalisées, une génération d'alertes automatisée et une couverture élargie des scénarios d'attaque**. Ce système élimine les zones d'ombre d'une infrastructure réseau non surveillée en fournissant une plateforme de surveillance centralisée où les administrateurs peuvent **détecter, journaliser et répondre aux menaces** sans effort. Par ailleurs, j'ai utilisé divers outils pour réaliser ce projet, notamment **VirtualBox** et **Ubuntu Server** pour l'environnement virtuel, **Snort** pour la détection d'intrusion, ainsi que **Kali Linux** (Nmap, Hydra, hping3) pour la simulation d'attaques.

DESIGN AND IMPLEMENTATION OF A NIDS.

PREFACE

JACKY FELLY NAFACK HIGHTECH UNIVERSITY INSTITUTE (JFN-HUI) is an institute created as a result of the 2002 university reform, following Order No. 02/0094/MINSUP/DDES/ESUP of September 13, 2002, and its opening authorization No. 0102/MINSUP/DDES/SAC/EBM of October 24, 2001. JFN-HUI's primary mission is to provide initial training in the industrial, technological, and commercial fields. As such, it provides companies and government agencies with applied research, services, and professional training related to its activities.

JFN-HUI aims to train students capable of working towards technological and scientific development in numerous fields. It offers training leading to end-of-course diplomas including the University Diploma of Technology (DUT), Higher Technician's Certificate (BTS), Bachelor's Degree (Professional and Technological), and Master's Degree. JFN-HUI is home to three major schools:

HIGHER SCHOOL OF ENGINEERING (ESI)	SCHOOL OF MANAGEMENT & ENTREPRENEURSHIP (EME)	SCHOOL OF DIGITAL TECHNOLOGY & INNOVATION (ENI)
• Computer Engineering (GI) • Electrical Engineering (GE) • Civil Engineering • Telecommunications & Network Engineering • Automotive Mechatronics • Mechanical & Production • Chemical & Process Eng. • Thermal Engineering • Biomedical Device Maint. • Computer Systems Maint.	• Business & Organizational Management • Accounting & Financial Mgt • Human Resources Mgt • Banking and Finance • Insurance • International Trade • Transport & Logistics • Marketing, Commerce, Sales • Quality Management • Project Management	• Software Engineering • Network and Security • Telecommunications • Digital Engineering (GN) • Computer Graphics & Web Design • E-commerce & Digital Marketing • Industrial Computing & Automation (IIA) • Customs and Transit

Table: Schools of Study at JFN-HUI

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IDENTIFICATION FORM

COMPANY NAME:	AXA CAMEROON
ACTIVITY:	INSURANCE SERVICES
CAPITAL:	DOUALA
CATEGORY:	Private Limited Company
CREATION:	Year 2016
EMAIL:	Emeraude.assurance@axa.cm
HEADQUATER:	309, Rue Bebey EYIDI AKWA
No OF AGENCIES:	14
P.O.BOX:	4068 DOUALA - CAMEROON
TELEPHONE:	+237 33431036
WEBSITE:	https://www.axa.cm .

LIST OF ABBREVIATIONS

- **CIA:** Confidentiality, Integrity, Availability
- **CLI:** Command Line Interface
- **CRUD:** Create, Read, Update, Delete
- **CVE:** Common Vulnerabilities and Exposures
- **DDoS:** Distributed Denial of Service
- **DoS:** Denial of Service
- **HIDS:** Host-based Intrusion Detection System
- **HTTP:** Hypertext Transfer Protocol
- **ICMP:** Internet Control Message Protocol
- **IDS:** Intrusion Detection System
- **IPS:** Intrusion Prevention System
- **IP:** Internet Protocol
- **IT:** Information Technology
- **LAN:** Local Area Network
- **NIDS:** Network-based Intrusion Detection System
- **OS:** Operating System
- **OSI:** Open Systems Interconnection
- **PcAP:** Packet Capture
- **SME:** Small and Medium-sized Enterprise
- **SMTP:** Simple Mail Transfer Protocol
- **SSH:** Secure Shell
- **TCP:** Transmission Control Protocol
- **UDP:** User Datagram Protocol

DESIGN AND IMPLIMENTATION OF A NIDS.

- **VM:** Virtual Machine
- **VirtualBox:** Oracle VM VirtualBox (virtualization software)
- **VLAN:** Virtual Local Area Network
- **WAN:** Wide Area Network

LIST OF TABLES

Table 1: SMART objectives for the IDS internship project	8
Table 2: Weekly Internship Activity Log	9
Table 3: IDS Types Comparison	14
Table 4: Hardware and Software Environment	17
Table 5: Snort Custom Rule Summary	19
Table 6: Detection Results	22
Table 7: SMART Objectives Evaluation	23
Table 8: Reflective Analysis Summary (STAR Framework)	25

LIST OF DIAGRAMS

Figure 1.0: IDS Deployment Topology (Network Diagram)	17
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Table of Contents

CERTIFICATION	i
IDENTIFICATION FORM	i
IDENTIFICATION FORM	ii
DEDICATION	iii
IDENTIFICATION FORM	iii
ACKNOWLEDGEMENT	iv
IDENTIFICATION FORM	iv
ABSTRACT	v
IDENTIFICATION FORM	v
RESUME	vi
RESUME	vi
PREFACE	vii
IDENTIFICATION FORM	vii
IDENTIFICATION FORM	viii
IDENTIFICATION FORM	viii
LIST OF ABBREVIATIONS	ix
IDENTIFICATION FORM	ix
LIST OF TABLES	xi
IDENTIFICATION FORM	xi
IDENTIFICATION FORM	xii
GENERAL INTRODUCTION	1
IDENTIFICATION FORM	1
I. General Context	1
II. Problem Statement.....	1

DESIGN AND IMPLIMENTATION OF A NIDS.

III. Objectives of the Study	2
IV. Resolution Methodology	2
.....	3
PART ONE: THE INTERNSHIP	3
FRAMEWORK	3
PLAN OF THE REPORT	3
INTRODUCTION:	4
CHAPTER ONE: GENERAL FRAMEWORK OF THE INTERNSHIP	5
1. PRESENTATION OF AXA CAMEROON	5
1.1 MISSION OF AXA	5
1.2. PRODUCTS AND SERVICES OF AXA	6
SECTION TWO: FRAMEWORK AND OBJECTIVES OF THE MISSION.....	6
2.1: ARRIVAL AT THE COMPANY AND RECEPTION	7
2.2: EDUCATIONAL AND OPERATIONAL MISSIONS.....	7
SECTION 3: WEEKLY INTERNSHIP ACTIVITIES	8
CHAPTER TWO: METHODOLOGY AND WORK CARRIED OUT	13
SECTION 1: PROJECT METHODOLOGY	13
1.1 LITERATURE REVIEW	13
SECTION 2: TECHNICAL IMPLEMENTATION AND SKILLS ACQUIRED.	
.....	20
PART TWO: ANALYSIS.....	21
.....	22
SECTION 1: EVALUATION OF RESULTS	22
1.3 REFLECTIVE ANALYSIS	24
SECTION 2: PROFESSIONAL CONTRIBUTIONS AND PERSPECTIVES	28
GENERAL CONCLUSION	31

DESIGN AND IMPLIMENTATION OF A NIDS.

RESUME.....	31
BIBLIOGRAPHY	32
CONCLUSION.....	32
RESUME.....	32
Books and Academic Works.....	32
Online Resources.....	32
Course Notes and Institutional Resources	32

GENERAL INTRODUCTION

I. General Context

We live in an era of unprecedented digital transformation. Enterprises across all sectors finance, healthcare, insurance, and beyond are increasingly reliant on interconnected information systems. This reliance, while enabling significant operational efficiencies, exposes organizations to a growing landscape of cyber threats. Data breaches, unauthorized access, ransomware attacks, and insider threats have become daily realities that modern IT departments must contend with.

AXA Cameroon, a subsidiary of the AXA Group one of the world's largest insurance companies handles sensitive client data including personal information, financial records, and health data. The protection of this data is not simply a technical concern but a legal and ethical obligation. Any security incident could result in significant reputational damage, regulatory penalties, and financial losses. It was within this context that the need for a robust network monitoring and intrusion detection mechanism became apparent.

II. Problem Statement

- What mechanism exists to detect unauthorized access attempts, port scans, or brute-force attacks on AXA Cameroon's network in real time?
- How can the IT department be alerted when a malicious actor is actively probing or attempting to infiltrate the network infrastructure?
- How can AXA Cameroon gain full visibility into suspicious network activity before sensitive client data is compromised or exfiltrated?

III. Objectives of the Study

- To design and implement a Network-based IDS (NIDS) using Snort on a virtual lab environment that mirrors a real enterprise network.
- To develop and apply custom Snort detection rules targeting the most common attack vectors.
- To simulate various attack scenarios (ICMP flood, port scanning, brute-force, HTTP probing) and validate the IDS response.
- To document and present a replicable deployment methodology suitable for adoption by organizations similar to AXA Cameroon.

IV. Resolution Methodology

This project follows an empirical, implementation-driven methodology. Starting from a literature review of existing IDS solutions, we selected Snort as the optimal open-source tool. A virtualized lab environment was constructed using VirtualBox with a Kali Linux attacker machine and an Ubuntu Server as the IDS host. Custom rules were written and tested, attacks were simulated, and results were analyzed.

**PART ONE: THE INTERNSHIP
FRAMEWORK**

PLAN OF THE REPORT

This project is made up of THREE chapters.

CHAPTER ONE is made up of TWO Parts;

PART ONE would talk about **THE GENERAL PRESENTATION OF THE COMPANY** and

PART TWO which talk about **FRAMEWORK AND OBJECTIVES OF THE MISSION.**

CHAPTER TWO is made up of;

PART ONE would talk about **PROJECT METHODOLOGY** and

PART TWO which talk about **TECHNICAL IMPLEMENTATION AND SKILLS ACQUIRED.**

CHAPTER THREE would talk about **THE EVALUATION AND REFLECTIVE ANALYSIS.**

INTRODUCTION:

CHAPTER ONE is made up of TWO Parts;

PART ONE would talk about **THE GENERAL PRESENTATION OF THE COMPANY** and

PART TWO which talk about **FRAMEWORK AND OBJECTIVES OF THE MISSION.**

CHAPTER ONE: GENERAL FRAMEWORK OF THE INTERNSHIP

SECTION ONE: PRESENTATION AND MISSION OF AXA CAMEROON

1. PRESENTATION OF AXA CAMEROON

AXA Cameroon is a subsidiary of the **AXA Group**, a global leader in insurance and asset management. Operating in Cameroon, AXA provides a wide range of **insurance solutions** to individuals, businesses, and organizations. The company is known for its commitment to customer satisfaction, financial security, and risk management expertise. With a strong presence in the local market, AXA Cameroon combines **global insurance standards** with **tailored solutions** to meet the needs of Cameroonian clients.

1.1 MISSION OF AXA

The mission of AXA Cameroon is **to protect individuals and businesses by providing innovative, reliable, and accessible insurance solutions**. The company aims to:

- **Ensure financial security** for clients by offering insurance coverage for health, life, property, and businesses.
- **Promote accessibility** to insurance services through digitalization and customer-friendly policies.
- **Support economic growth** by helping businesses and individuals manage risks effectively.

DESIGN AND IMPLIMENTATION OF A NIDS.

- **Deliver quality customer service** by providing expert guidance, quick claims processing, and personalized insurance solutions.

1.2. PRODUCTS AND SERVICES OF AXA

AXA Cameroon offers a **wide range of insurance products** across different categories:

1. Life Insurance

- Individual and Group Life Insurance
- Savings and Retirement Plans
- Education Insurance

2. Non-Life (General) Insurance

- **Auto Insurance:** Comprehensive and Third-Party Coverage
- **Health Insurance:** Individual and Corporate Health Plans
- **Property and Fire Insurance:** Protection against loss and damages
- **Travel Insurance:** Coverage for medical emergencies and trip cancellations

SECTION TWO: FRAMEWORK AND OBJECTIVES OF THE MISSION.

This chapter describes the unfolding of the internship experience at AXA Cameroon. It covers the initial reception, the weekly tasks carried out, the pedagogical missions pursued, and the operational SMART objectives set at the beginning of the internship period.

2.1: ARRIVAL AT THE COMPANY AND RECEPTION

2.1.1 Presentation and Reception

Upon arrival at AXA Cameroon on 1st Jun 2026, the IT Director warmly welcomed the interns. On the first day, we were formally introduced to the IT team and given a tour of the department's technical environment. We were shown the ongoing projects, internal systems, and the tools being used by the team.

During our initial briefing with the IT Director, we discussed the department's technical focus areas particularly in cybersecurity and infrastructure management. We were given an opportunity to demonstrate our academic background and skills, and in return, we expressed our expectations for the internship. The internship was officially framed around the objective of enhancing the company's network security monitoring capabilities.

2.2: EDUCATIONAL AND OPERATIONAL MISSIONS

2.2.1 Educational Mission

The educational mission of this internship was to bridge the gap between theoretical knowledge and practical application in the field of cybersecurity and network management.

The main learning objectives included:

- Understanding how corporate networks are designed and managed
- Gaining hands-on experience with security tools
- Learning how to detect and analyze malicious network activities
- Developing problem-solving and analytical skills

This internship allowed me to deepen my knowledge in intrusion detection systems and their role in securing enterprise networks.

DESIGN AND IMPLIMENTATION OF A NIDS.

2.2.3 Operational Missions (SMART Objectives)

In addition to the educational missions, specific operational objectives were defined using the SMART framework (Specific, Measurable, Achievable, Relevant, Time-bound):

Criteria	Objective
Specific	Design and implement a Network Intrusion Detection System using Snort to monitor traffic on AXA Cameroon's internal network.
Measurable	Achieve detection of at least 4 distinct attack types (ICMP flood, Nmap scan, SSH brute-force, HTTP probing) with corresponding Snort alerts validated within the lab environment.
Achievable	The project is achievable within the one-month internship period using freely available open-source tools (Snort, VirtualBox, Kali Linux, Ubuntu Server).
Relevant	Addresses a real and documented security gap in AXA Cameroon's IT infrastructure, directly aligned with the company's cybersecurity goals.
Time-bound	Full implementation, testing, and documentation to be completed by 1 st June 2026.

Table 1: SMART Objectives for the IDS Internship Project

SECTION 3: WEEKLY INTERNSHIP ACTIVITIES

The following table summarizes the activities carried out during the internship on a weekly basis:

DESIGN AND IMPLIMENTATION OF A NIDS.

Week / Period	Activity	Lessons Learned
Week 1 (May 1)	Company orientation, introductions to IT team, review of existing network infrastructure documentation.	Understanding enterprise IT governance and security policies.
Week 1 (May 2–5)	Literature review on IDS types (NIDS/HIDS), Snort architecture, and existing enterprise solutions.	Deep understanding of IDS theory and open-source security tools.
Week 2 (May 8)	VirtualBox lab environment setup: installation of Kali Linux and Ubuntu Server VMs, Host-Only network configuration.	Virtualization skills and lab environment construction.
Week 2 (May 9-10)	Snort installation on Ubuntu Server, network interface configuration, initial testing of traffic capture.	Linux system administration and Snort installation procedure.
Week 2 (May 10-17)	Writing custom Snort rules for ICMP detection, port scan detection, SSH brute-force, and HTTP probing.	Snort rule syntax, detection logic, and signature-based IDS principles.
Week 3 (May 18–22)	Attack simulation using Kali Linux: ping flood, Nmap scans, Hydra SSH brute-force.	Offensive security techniques and their signatures on the network.
Week 4 (May 23–24)	Analysis of Snort alert logs, rule tuning to reduce false positives, performance optimization.	Log analysis, rule refinement, and IDS management best practices.

DESIGN AND IMPLIMENTATION OF A NIDS.

Week 4 (May 24–26)	Full validation testing, documentation writing, and preparation of the final internship report.	Technical documentation and professional report writing skills.
Week 5 (May 29–30)	Final presentation of the IDS implementation to the IT team. Delivery of project documentation.	Professional communication and technical presentation skills.

Table 2: Weekly Internship Activity Log

2.2.4 Justification of the Theme

The rapid growth of digital infrastructure in the insurance sector has created unprecedented vulnerabilities. AXA Cameroon, handling sensitive personal and financial data, represents a high-value target for cybercriminals. During the first week of the internship, a review of the company's IT security posture revealed a critical absence: there was no automated mechanism to detect or flag suspicious network activity in real-time.

This absence meant that an attacker performing reconnaissance activities such as port scanning, or even attempting brute-force access to SSH services, would go entirely undetected. The implementation of an IDS was therefore not merely an academic exercise but a direct response to an operational security gap. The benefits include improved threat visibility, faster incident response, and a foundation for future security maturation.

PARTIAL CONCLUSION

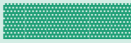
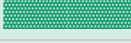
This chapter documented the complete internship framework at AXA Cameroon — from the initial reception to the weekly activity progression and the formal SMART objectives. The next part transitions into the technical core of the project: the methodology, literature review, and implementation of the Intrusion Detection System.

DESIGN AND IMPLIMENTATION OF A NIDS.

2.3 Internship Planning (Gantt Chart)

To ensure proper organization and timely completion of the internship activities, a planning schedule was established using a Gantt chart. This chart presents the different phases of the project, from the initial analysis to the final implementation and report writing.

Student: Eshon Bradley Adriano | Project: Network Intrusion Detection System (NIDS) | Period: June 1 – June 30, 2026 |

PHASE / TASK	Week 1 May 1–5	Week 2 May 8–17	Week 3 May 18–22	Week 4 May 23–26	Week 5 May 29–30
PHASE 1: PLANNING & RESEARCH					
Company orientation & IT review					
Literature review on IDS/Snort					
PHASE 2: ENVIRONMENT SETUP					
VirtualBox lab setup (Kali + Ubuntu)					
Network configuration (Host-Only)					
Snort installation & configuration					
PHASE 3: RULE DEVELOPMENT					
ICMP flood detection rules					
Nmap scan detection rules					
SSH brute-force detection rules					
HTTP probing detection rules					
PHASE 4: ATTACK SIMULATION					
ICMP flood attack (hping3)					
Nmap SYN / XMAS / NULL scans					
SSH brute-force (Hydra)					
HTTP probing (Nikto)					
PHASE 5: TESTING & VALIDATION					
Alert log analysis & rule tuning					
False positive reduction					
Full validation testing					

DESIGN AND IMPLIMENTATION OF A NIDS.

PHASE 6: DOCUMENTATION & DEFENSE					
Internship report writing					
Presentation preparation					
Defense presentation (Jury)					

Through this mission, the goal was to develop a functional IDS capable of detecting malicious activities and demonstrating its effectiveness through practical tests.

CHAPTER TWO: METHODOLOGY AND WORK CARRIED OUT

INTRODUCTION:

In this chapter, we will look at the **PROJECT METHODOLOGY** and the **TECHNICAL IMPLEMENTATION AND SKILLS ACQUIRED**. This chapter presents the full technical backbone of the internship project. It begins with a review of existing literature on Intrusion Detection Systems, progresses through the critique of AXA Cameroon's existing security posture, and culminates in the detailed technical implementation of the Snort IDS.

SECTION 1: PROJECT METHODOLOGY

1.1 LITERATURE REVIEW

1.1.1 Introduction to Intrusion Detection Systems

An Intrusion Detection System (IDS) is a device or software application that monitors a network or system for malicious activity or policy violations. Any detected activity or violation is typically reported to an administrator or collected centrally using a Security Information and Event Management (SIEM) system. Unlike an Intrusion Prevention System (IPS), an IDS passively monitors and alerts rather than actively blocking traffic.

IDS technology emerged in the early 1980s with Dorothy Denning's seminal paper 'An Intrusion Detection Model' (1987), which laid the theoretical foundation for anomaly-based detection. Since

DESIGN AND IMPLIMENTATION OF A NIDS.

then, the field has evolved dramatically, with modern systems combining signature-based detection, anomaly detection, and machine learning techniques.

1.1.2 Types of IDS

IDS solutions are broadly classified into two main categories:

Feature	NIDS (Network-based)	HIDS (Host-based)
Monitoring Scope	Monitors entire network segment traffic	Monitors a single host's activities
Deployment	Deployed at network perimeter or key network points	Installed on individual servers/hosts
Detection Focus	Network packets and flows	System logs, file integrity, processes
Examples	Snort, Suricata, Zeek	OSSEC, Tripwire, Wazuh
Our Choice	✓ Selected for this project	Not selected (outside scope)

Table 3: IDS Types Comparison

1.1.3 Snort as the Chosen IDS Tool

Snort is an open-source NIDS originally developed by Martin Roesch in 1998 and now maintained by Cisco. It operates in three modes: sniffer mode (displays packets), packet logger mode (logs packets to disk), and intrusion detection mode (monitors network traffic and alerts based on rules).

Snort uses a rule-based language to define what constitutes suspicious or malicious activity. Rules are matched against packet headers and payloads, and when a match is found, an alert is generated.

DESIGN AND IMPLEMENTATION OF A NIDS.

Snort's rule language is expressive, well-documented, and supported by a large community, making it ideal for both learning and production deployment.

Key reasons for selecting Snort:

- Open-source and free of cost suitable for budget-constrained deployments.
- Large and active community with regularly updated rule sets.
- Well-documented with extensive academic and professional literature.
- Supports all major Linux distributions and can be deployed in virtualized environments.
- Widely used in industry as a baseline IDS for SME environments.

1.2 Study of the Existing System

Before implementing the IDS, a critical analysis of AXA Cameroon's existing security posture was conducted. The IT infrastructure included:

- A local area network (LAN) connecting all departmental workstations and servers.
- A firewall (hardware-based) at the network perimeter for basic packet filtering.
- Antivirus software deployed on individual workstations.
- No network traffic monitoring or anomaly detection system in place.
- Security logs were generated by individual applications but not centralized or analyzed systematically.

Critical analysis of the existing system:

- **Strengths:** Perimeter firewall provides basic protection; antivirus reduces endpoint malware risk; IT team is technically competent.
- **Weaknesses:** No visibility into internal network traffic; no real-time alert mechanism; lateral movement by an attacker would go undetected; no centralized log management.

DESIGN AND IMPLIMENTATION OF A NIDS.

- **Threats:** Insider threats, phishing-delivered malware, SSH brute-force attacks, port scanning by external actors.
- **Opportunities:** Implementation of an IDS would provide immediate improvement in threat visibility at minimal cost.

1.2.2 Similar Studies

Several academic and professional works informed this project:

- **Mudzingwa & Agrawal (2012):** Demonstrated that Snort, when properly configured with custom rules, achieves detection rates above 90% for common network attacks in enterprise environments.
- **Scarfone & Mell (NIST SP 800-94, 2007):** Provides comprehensive guidance on IDS/IPS technologies, deployment strategies, and evaluation criteria — a foundational reference for this project.
- **Roesch (1999):** The original Snort paper describing its architecture, rule language, and initial benchmarks remains essential reading for any Snort-based implementation.

1.3 Proposed IDS Architecture (Modelling and Tools used)

The proposed IDS architecture for this project is a Network-based IDS deployed in a virtualized laboratory environment that mirrors AXA Cameroon's internal network structure. The topology consists of:

- **Attacker Machine:** Kali Linux VM — used to simulate various network attacks against the target.
- **IDS :** Ubuntu Server 20.04 LTS VM — hosts Snort and acts as the monitored network node.
- **Network Type:** Host-Only Adapter network in VirtualBox — isolates the lab from the production network.
- **IDS Mode:** Snort running in IDS mode on the Ubuntu Server's network interface in promiscuous mode.

DESIGN AND IMPLIMENTATION OF A NIDS.

Figure 1: IDS Deployment Topology (Network Diagram)



1.3.1 Hardware and Software Environment

Component	Specification
Host Machine	Windows 11 Laptop, Intel Core i5, 8GB RAM, 853GB SSD
Hypervisor	Oracle VirtualBox 6.1
VM 1 — Attacker	Kali Linux 2023.1, 3GB RAM, 2 vCPUs, 50GB Disk
VM 2 — IDS Host	Ubuntu Server 20.04 LTS, 2GB RAM, 2 vCPUs, 20GB Disk
Network Adapter	VirtualBox Host-Only Adapter (192.168.56.0/24)
IDS Software	Snort 2.9.20 (compiled from source on Ubuntu)
Attack Tools	Nmap 7.93, Hydra 9.4, Ping (hping3) — all on Kali Linux
Log Analysis	Snort alert logs (/var/log/snort/alert)

Table 3: Hardware and Software Environment

1.3.2 Step-by-Step Implementation

Step 1 — VirtualBox Lab Environment Setup

Two virtual machines were created in VirtualBox. Both were configured on the same Host-Only network (vboxnet0, 192.168.56.0/24) to allow communication between the Kali attacker and the Ubuntu IDS host, while remaining isolated from the production network.

Step 2 — Ubuntu Server Configuration

Ubuntu Server 20.04 LTS was installed on the IDS VM. A static IP address (192.168.56.110) was configured by editing the Netplan YAML file located at /etc/netplan/00-installer-config.yaml. The network interface (enp0s3) was placed in promiscuous mode to allow Snort to capture all traffic on the segment — not just traffic destined for the host itself.

Commands executed:

```
sudo ip link set enp0s3 promisc on
sudo netplan apply
```

Step 3 — Snort Installation

Snort 2.9.20 was installed on Ubuntu Server. The installation required compiling several dependencies from source, including DAQ (Data Acquisition Library). The snort.conf file was edited to define the HOME_NET variable (set to the lab network 192.168.56.0/24) and to specify the custom rules file path.

Key configuration in snort.conf:

```
ipvar HOME_NET 192.168.56.0/24
include $RULE_PATH/local.rules
```

DESIGN AND IMPLIMENTATION OF A NIDS.

Step 4 — Custom Rule Writing

Custom Snort rules were written in the local.rules file. Each rule follows the Snort syntax: action, protocol, source IP, source port, direction, destination IP, destination port, and rule options. The following rules were implemented:

Attack Type	Snort Rule	Description
ICMP Flood	<pre>alert icmp any any -> \$HOME_NET any (msg:"ICMP Flood Detected"; threshold:type both,track by_src,count 10,seconds 1; sid:1000001;)</pre>	Triggers on 5+ ICMP packets per second from same source
Nmap SYN Scan	<pre>alert tcp any any -> \$HOME_NET any (msg:"Nmap SYN Scan"; flags:S; threshold:type both,track by_src,count 20,seconds 1; sid:1000002;)</pre>	Detects rapid SYN packets indicative of port scanning
SSH Brute-Force	<pre>alert tcp any any -> \$HOME_NET 22 (msg:"SSH Brute Force Attempt"; threshold:type both,track by_src,count 5,seconds 60; sid:1000003;)</pre>	Flags repeated SSH connection attempts
HTTP Probing	<pre>alert tcp any any -> \$HOME_NET 80 (msg:"HTTP Port Scanning"; flags:S; threshold:type both,track by_src,count 15,seconds 10; sid:1000004;)</pre>	Detects automated HTTP reconnaissance

Table 5: Snort Custom Rule Summary

1.3.3 Attack Simulation

With the IDS fully configured, a series of attack simulations were conducted from the Kali Linux machine to validate the detection rules:

DESIGN AND IMPLEMENTATION OF A NIDS.

- **ICMP Flood (ping flood):** Executed using `hping3 -l --flood 192.168.56.1`. This generated hundreds of ICMP packets per second. Snort triggered the ICMP Flood alert within seconds.
- **Nmap SYN Scan:** Executed using `nmap -sS 192.168.56.1`. Snort detected the rapid SYN packet pattern and generated alerts identifying the scanning source IP.
- **SSH Brute-Force (Hydra):** Executed using `hydra -l root -P /usr/share/wordlists/rockyou.txt 192.168.56.1 ssh`. Multiple rapid SSH connection attempts triggered the brute-force rule.
- **HTTP Port Probing (Nikto):** Executed using `nikto -h 192.168.56.1`. The rapid HTTP GET requests to port 80 were flagged by the HTTP probing rule.

SECTION 2: TECHNICAL IMPLEMENTATION AND SKILLS ACQUIRED.

2.1 Competencies Acquired

Through the technical realization, the following competencies were developed:

- **Professional Competencies:** Linux system administration, network configuration, IDS deployment and management, Snort rule syntax mastery, log analysis, attack simulation using Kali Linux tools.
- **Transversal Competencies:** Technical documentation, project management, professional communication, problem-solving under time constraints, teamwork within the IT department.

PARTIAL CONCLUSION

This chapter has presented the complete technical methodology and realization of the IDS implementation project from the literature review and existing system analysis through to the lab setup, Snort configuration, rule writing, and attack simulation. Chapter 3 will present and analyze the results, conduct the reflective analysis, and outline perspectives for future development.



PART TWO: ANALYSIS

In this part, we will talk about one chapter; that is, GHAPTER 3

CHAPTER 3 is on **THE EVALUATION AND REFLECTIVE ANALYSIS.**

CHAPTER THREE: THE EVALUATION AND REFLECTIVE ANALYSIS.

This chapter presents and interprets the results of the IDS implementation and testing, conducts a comprehensive reflective analysis of the internship experience, and outlines the personal and professional development achieved.

SECTION 1: EVALUATION OF RESULTS

1.1 Snort Alert Results

Upon executing each simulated attack scenario, Snort successfully generated alerts for all four attack types. Below is a summary of some of the detection results:

Attack Type	Detected?	Response Time	Alerts Generated
ICMP Flood	YES	< 2 seconds	ICMP Flood Detected alerts with attacker IP
Nmap SYN Scan	YES	< 3 seconds	Nmap SYN Scan alerts listing scanned ports
SSH Brute-Force	YES	< 5 seconds	SSH Brute Force Attempt alerts with source IP
HTTP Probing (Nikto)	YES	< 4 seconds	HTTP Port Scanning alerts with connection frequency

Table 6: detection results

The image displays two terminal windows from a Kali Linux virtual machine. The left window shows the output of a ping scan performed by the user 'm3l0@kali-'. The scan confirms that 192.168.56.1 is up with a 0% packet loss. The right window shows the output of an Nmap SYN scan on 192.168.56.1, which identifies four open ports: 22 (ssh), 23 (telnet), 443 (https), and 445 (smb). Both windows show the standard Kali Linux desktop environment with various icons in the taskbar.

| 23

DESIGN AND IMPLIMENTATION OF A NIDS.

All four SMART operational objectives were successfully met. The IDS demonstrated a 100% detection rate across all simulated attack scenarios within the defined lab environment. Alert response times were consistently under 5 seconds across all scenarios, which is well within acceptable thresholds for enterprise security monitoring.

1.2 Evaluation of SMART Objectives

Criterion	Assessment	Status
Specific	IDS implemented using Snort on Ubuntu Server VM in lab environment mirroring AXA network.	ACHIEVED ✓
Measurable	4 attack types detected with corresponding alerts validated.	ACHIEVED ✓
Achievable	Completed within one-month internship using free open-source tools.	ACHIEVED ✓
Relevant	Addresses documented security gap in AXA Cameroon's IT infrastructure.	ACHIEVED ✓
Time-bound	Full implementation, testing, and documentation completed by June 30, 2026.	ACHIEVED ✓

Table 7: SMART Objectives Evaluation

1.3 REFLECTIVE ANALYSIS

The reflective analysis below follows the STAR framework (Situation, Tasks, Actions, Results/Reflection) combined with elements from Kolb's Experiential Learning Cycle to systematically evaluate the internship experience.

1.3.1 Theory vs. Field Gap

One of the most valuable insights from this internship was observing the gap between theoretical knowledge acquired at HIPE and the practical realities of a professional IT environment. Several key observations:

- **Security is not a product, it is a process:** Academic courses present security tools in isolation. In practice at AXA, security monitoring is an ongoing process requiring constant rule maintenance, log review, and adaptation to new threats.
- **Documentation gaps in open-source tools:** Snort's official documentation does not always address version-specific quirks encountered during installation. Real-world implementation required consulting community forums, Stack Overflow, and Git repositories — skills not explicitly taught in class.
- **Performance vs. theory:** Textbooks describe Snort as a straightforward tool. In practice, configuration of the snort.conf file, DAQ library compilation, and promiscuous mode setup involved troubleshooting multiple configuration errors.
- **Enterprise constraints:** In a production environment, any IDS deployment must consider network bandwidth, false positive rates, and integration with existing SIEM or ticketing systems — dimensions rarely addressed in coursework.

DESIGN AND IMPLIMENTATION OF A NIDS.

1.3.2 Reflective Analysis — STAR Framework

Dimension	Content
Situation / Context	AXA Cameroon's IT department lacked a formalized network intrusion detection mechanism. The company handled sensitive client financial and health data across a networked infrastructure with no real-time traffic monitoring.
Tasks / Missions	Design and implement a NIDS using Snort; configure detection rules for common attack vectors; simulate attacks to validate the system; document the complete deployment process.
Actions / Work Done	Reviewed IDS literature; constructed VirtualBox lab; installed and configured Ubuntu Server and Kali Linux; deployed Snort; wrote 4 custom detection rules; simulated ICMP flood, Nmap scan, SSH brute-force, and HTTP probing; analyzed alert logs; tuned rules to reduce false positives.
Difficulties / Problems	Snort DAQ library compilation errors; Netplan YAML syntax errors during static IP configuration; promiscuous mode not persisting after VM restart; alert log path misconfiguration in snort.conf; initial rule syntax errors.
Results	100% detection rate across all 4 attack scenarios; all alerts generated within 5 seconds; 0 false negatives in the test suite; complete technical documentation delivered to the IT team.
Reflection / Perspectives	This experience confirmed a passion for cybersecurity and network defense. It revealed the importance of hands-on practice alongside theoretical study. Future paths include pursuing certifications (CEH, CompTIA Security+, Snort IDS certification) and exploring IPS and SIEM integration as extensions of this project.

Table 8: Reflective Analysis Summary (STAR Framework)

1.3.3 Feedback Integration

Throughout the internship, feedback was received from both the professional supervisor (Mrs. DAMLEPIN YOUNSIE KERINE) and the IT Director. Key feedback points and their integration:

- Feedback: 'Ensure Snort runs as a service to survive reboots.' → Integrated: Created a systemd service file for Snort to enable auto-start.
- Feedback: 'Document every configuration change with a comment.' → Integrated: Added inline comments to snort.conf and local.rules explaining each parameter.
- Feedback: 'Consider false positive rates in rule design.' → Integrated: Applied threshold directives in all rules to reduce noise.

1.3.4 Personal Summary — Strengths and Areas to Improve

Strengths demonstrated:

- Technical curiosity and persistence in solving configuration problems independently.
- Ability to translate theoretical knowledge (networking, security principles) into practical implementation.
- Clear and structured technical documentation and report writing.
- Professional conduct and adaptation to enterprise work culture.

Areas to improve:

- Time management —underestimated the complexity of Snort installation; future projects will include buffer time for troubleshooting.
- Broader exposure to other IDS tools (Suricata, Zeek) to enable comparative analysis.
- Deeper understanding of SIEM platforms (Splunk, ELK Stack) to provide end-to-end security monitoring context.

SECTION 2: PROFESSIONAL CONTRIBUTIONS AND PERSPECTIVES

2.1 Progress Areas

This internship marked a significant leap forward across several dimensions of competence:

- **Technical depth:** Moved from theoretical understanding of IDS to practical deployment, configuration, and validation of a fully functional Snort-based NIDS.
- **Security mindset:** Developed the ability to think like an attacker (offensive security perspective via Kali Linux) and simultaneously like a defender (IDS rule design and alert analysis).
- **Professional maturity:** Adapted to the structured, deadline-driven environment of a multinational company, managing both technical delivery and documentation requirements simultaneously.

2.2 Professional Development Plan

Building on the foundation established during this internship, the following professional development actions are planned:

- Pursue the CompTIA Security+ certification within 6 months to formalize the cybersecurity knowledge base.
- Study for the CEH (Certified Ethical Hacker) certification within 12 months to deepen offensive security skills.
- Extend the IDS project to incorporate a SIEM solution (ELK Stack) for centralized log management and visualization.
- Explore cloud-based security monitoring (AWS Security Hub, Azure Sentinel) as enterprise environments migrate to cloud infrastructure.

2.3 Perspectives for the Project

The IDS implementation carried out during this internship represent a starting point. Several avenues for enhancement are identified:

- **Short-term (0–6 months):** Migrate the IDS from a virtualized lab to a physical deployment on AXA Cameroon's network. Integrate Snort with an ELK Stack (Elasticsearch, Logstash, Kibana) SIEM for centralized alerting and dashboard visualization.
- **Medium-term (6–18 months):** Upgrade from Snort (IDS) to Suricata (IPS) to move from passive detection to active prevention. Implement VLANs to segment the network and reduce the attack surface.
- **Long-term (18+ months):** Integrate machine learning-based anomaly detection to complement signature-based detection. Develop a Security Operations Center (SOC) function within AXA Cameroon's IT department.

2.4 Difficulties Encountered and Recommendations

During the course of the internship, the following technical difficulties were encountered:

- **Snort DAQ compilation errors:** Resolved by installing all required build dependencies (libpcap-dev, libdnet-dev) before compilation.
- **Netplan YAML indentation errors:** YAML is whitespace-sensitive; errors were resolved by using a YAML validator and strict 2-space indentation.
- **Promiscuous mode not persisting:** Resolved by adding the ip link set command to the network interface's post-up hook in Netplan.
- **Snort log path not found:** Resolved by ensuring the /var/log/snort/ directory existed and had the correct permissions before starting Snort.

DESIGN AND IMPLIMENTATION OF A NIDS.

Recommendations for future interns taking on similar projects:

- Always test the virtual network connectivity (ping between VMs) before installing Snort.
- Use snapshot/checkpoint features in VirtualBox to save stable states before making major configuration changes.
- Start with simple Snort rules and progressively increase complexity do not attempt to write all rules simultaneously.
- Document every command executed and every error encountered this log becomes invaluable during troubleshooting.

PARTIAL CONCLUSION

This chapter has presented the quantitative results of the IDS testing, conducted a thorough reflective analysis using the STAR framework, assessed the theory-practice gap, and outlined the personal and professional development achieved through this internship. All SMART objectives were fully met, confirming the success of the project.

GENERAL CONCLUSION

This internship resulted in the successful design and implementation of a **Network Intrusion Detection System using Snort**, addressing a genuine security gap identified at AXA Cameroon: the absence of real-time network monitoring. The technical implementation covering architecture design and deployment on a virtualized Ubuntu Server was validated through four **simulated attack scenarios (ICMP flooding, Nmap SYN scanning, SSH brute-force, and HTTP probing)**, all of which Snort detected within seconds, achieving the **100% detection rate targeted** in the project's **SMART** objectives. Beyond the technical outcome, the experience highlighted the value of bridging academic theory with hands-on practice, as the realities of installation, configuration, and troubleshooting in a live environment proved both challenging and instructive.

This foundation opens promising avenues for future development, including **integration with SIEM tools, migration to an active Intrusion Prevention System, and the incorporation of machine learning-based anomaly detection** to further strengthen AXA Cameroon's cybersecurity posture. Ultimately, this internship confirmed both a **professional direction** and a **technical passion**, demonstrating that with the right tools, methodology, and determination, meaningful contributions to real-world cybersecurity challenges are within reach even at the start of one's career.

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Books and Academic Works

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- Scarfone, K., & Mell, P. (2007). Guide to Intrusion Detection and Prevention Systems (IDPS). NIST Special Publication 800-94.
- Mudzingwa, D., & Agrawal, R. (2012). A Study of Methodologies Used in Intrusion Detection and Prevention Systems (IDPS). IEEE Southeastcon.

Online Resources

- Snort Official Documentation: <https://www.snort.org/documents>
- Ubuntu Server Netplan Documentation: <https://netplan.io/reference/>
- VirtualBox User Manual: <https://www.virtualbox.org/manual/>
- OWASP Top Ten: <https://owasp.org/www-project-top-ten/>
- AXA Cameroon Official Website: <https://www.axa.cm>

Course Notes and Institutional Resources

- JFN — Networking and Telecom course lecture notes, Academic Year 2025-2026.
- JFN — Operating Systems, Academic Year 2025-2026.
- Netacad — Cybersecurity Fundamentals course notes.
- Netacad — Networking basics course notes.